

Title: DPT

Date:

Model

$$W[A_{ij}|0 \rightarrow 1] = \frac{\alpha}{N} + \beta(A^2)_{ij} \quad W[A_{ij}|1 \rightarrow 0] = r$$

平均度 $\frac{dz}{dt} = \alpha - rz + \beta(z^2 - \Delta)$

$$\frac{dz}{dt} = \alpha \left(1 - \frac{z}{N}\right) - rz + \frac{1}{N} \langle \sum_{ijk} A_{ik} A_{kj} (1 - A_{ij}) \rangle$$

$$\beta = \frac{1}{z^2 - \Delta}$$

Δ : 三元组 $\frac{d\Delta}{dt} = \beta(z^2 - \Delta) - 3r\Delta + O(\beta^3) \quad (\vec{1} - \vec{r})$

$$z(t) \sim \begin{cases} \text{constant} & (\beta \leq \frac{r^2}{4\alpha}, t \rightarrow \infty) \\ \frac{1}{t - t_c} & (\beta > \frac{r^2}{4\alpha}, t \rightarrow t_c) \end{cases}$$

After DPT

p : iso node, $(1-p(t))N$: complete graph

$$q(t) = (1-p(t))^2 \quad dp/dt = \alpha t, \quad p(t) = p_0 e^{-\alpha(t-t_c)}$$

adiabatic approx:

$$\frac{d\Delta}{dt} = 0, \quad \Delta = \frac{\beta z^2}{\beta + 3r}, \quad \beta' = \beta \left(1 - \frac{\beta}{\beta + 3r}\right)$$

after the DPT:

$$\text{core } (1-p(t))N, \quad q(t) = \frac{z}{N} = (1-p(t))^2$$

$$dp/dt = \alpha p, \quad p = e^{-\alpha(t-t_c)} p_0$$

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dynamics

 α : constant growth

 β : high order

 Δ : triangle

 z : $\langle k \rangle$

$$\frac{dz}{dt} = \alpha - rz + \beta(z^2 - \Delta)$$

$$W[A_{ij}(0 \rightarrow 1)] = \left[\frac{\alpha}{N} + \beta(A^2)_{ij} \right]$$

$$W[A_{ij}(1 \rightarrow 0)] = r$$

$$\begin{aligned} \frac{dz}{dt} &= \alpha \left(1 - \frac{z}{N} \right) - rz \\ &+ \frac{\beta}{N} \left(\sum_{ijk} A_{ik} A_{kj} (1 - A_{ij}) \right) z^2 - \Delta \end{aligned}$$

hyperbolic scaling

$$z(t) \sim \begin{cases} \text{constant} & (\beta \leq \frac{r^2}{4\alpha}, t \rightarrow \infty) \\ 1/(t - t_c) & (\beta > \frac{r^2}{4\alpha}, t \rightarrow t_c) \end{cases}$$

$$\begin{aligned} t_c &= w^{-1} [\text{acot}(2w) + \pi/2] \\ (w &= \sqrt{|\alpha\beta - r^2/4|}) \end{aligned}$$

$$\frac{dz}{dt} = \alpha - rz + \beta z^2$$

$$1/(\alpha - rz + \beta z^2) dz = dt$$

$$\frac{1}{pu^2 + q} du = dt$$

$$p = \beta, u = z - \frac{r}{2\beta}, q = \alpha - \frac{r^2}{4\beta}$$

$$\frac{1}{\sqrt{pq}} \text{atan} \sqrt{\frac{p}{q}} u = t + C$$

$$t=0, z=0: C = \frac{1}{\sqrt{\alpha\beta - \frac{r^2}{4}}} \text{atan} \frac{-r}{\sqrt{4\alpha\beta - r^2}}$$

$$w = \sqrt{pq} \quad C = w^{-1} \text{atan} -\frac{r}{2w}$$

$$\begin{aligned} t_c &= -C + \frac{\pi}{2} w^{-1} \\ &= w^{-1} \left(\frac{\pi}{2} + \text{acot} \frac{2w}{r} \right) \end{aligned}$$